

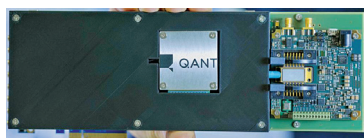
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

First Photonic Processor for Real-Time AI Applications

Q.ANT has launched its first product: a native processing unit (NPU) built on its LENA architecture (light empowered native arithmetics).



The NPU runs on a PCIe interface and works with existing computing environments. It performs non-linear calculations using light instead of electrons, offering at least 30x more energy efficiency and improved performance over CMOS processors. Made for applications like AI inference, machine learning, and physics simulations, the NPU has shown results in tasks such as number recognition in neural network inference.

Q.ANT

<http://qant.com>

Industry's Highest Memory Bandwidth FPGA

Altera Corporation has launched the Agilex 7 FPGA M-Series, the first FPGA



to combine high logic density with integrated high-bandwidth memory (HBM2E), DDR5, and LPDDR5 support. With over 3.8 million logic elements, it targets applications

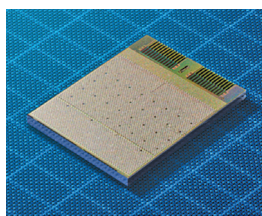
such as AI, data centres, firewalls, 5G infrastructure, and 8K broadcast systems. As AI, cloud services, and video streaming drive data growth, the need for more bandwidth, capacity, and power efficiency is increasing. Altera says the M-Series addresses these needs with fabric and memory interfaces that reduce latency and bottlenecks. It supports use cases including AI inference in data centres, memory buffers for firewall systems, data transfer for 8K video, and 800GbE test equipment.

Altera Corporation

<https://www.altera.com>

World's First UCle Optical Chiplet

Ayar Labs has introduced the Universal Chiplet Interconnect Express (UCle) optical chiplet. Designed to increase AI



infrastructure performance, the chiplet reduces latency and power use while

removing data bottlenecks. With a UCle electrical interface, it fits into existing chip designs. The TeraPHY optical I/O chiplet delivers up to 8Tbps of bandwidth, powered by Ayar Labs' 16-wavelength SuperNova light source. Its UCle interface allows interoperability across chiplets from different vendors, supporting a system for optical integration in AI systems. It targets OEMs, ODMs, and cloud providers building AI servers and infrastructure and telecom and networking companies needing data links across different distances.

Ayar Labs

<https://ayarlabs.com>

Industry's First High-Power LTE-M/NB-IoT Module

Ubiik has introduced the Maverick 220, an LTE-M/NB-IoT module designed for utilities, private LTE networks, and



industrial IoT. It delivers transmit power up to 28dBm for LTE-M and 30dBm for NB-IoT, exceeding the standard 23dBm limit and extending the range of LTE-connected IoT devices. This capability is useful when operating in narrowband channels (1 to 5MHz), often used in private LTE deployments. Ubiik states that the Maverick 220 is the first LTE module designed to address these limitations, improving performance in bandwidth-limited environments. It supports longer connectivity and up to 50% more range than standard LTE modules.

Ubiik

<https://www.ubiik.com>

World's First 12-Layer HBM4

SK Hynix has launched 12-layer HBM4, a DRAM designed for AI applications, for customers. The 12-layer HBM4 samples provide increased capacity and speed, with bandwidth capable of pro-



cessing over 2 terabytes of data per second—more than 60% faster

than the previous generation, HBM3E. This allows for processing over 400 full-HD movies every second. It is suited for